26–27 Data Logger Board (Hotfire Board)

Hardware Design Write-up

Gaucho Rocket Project

October 14, 2026

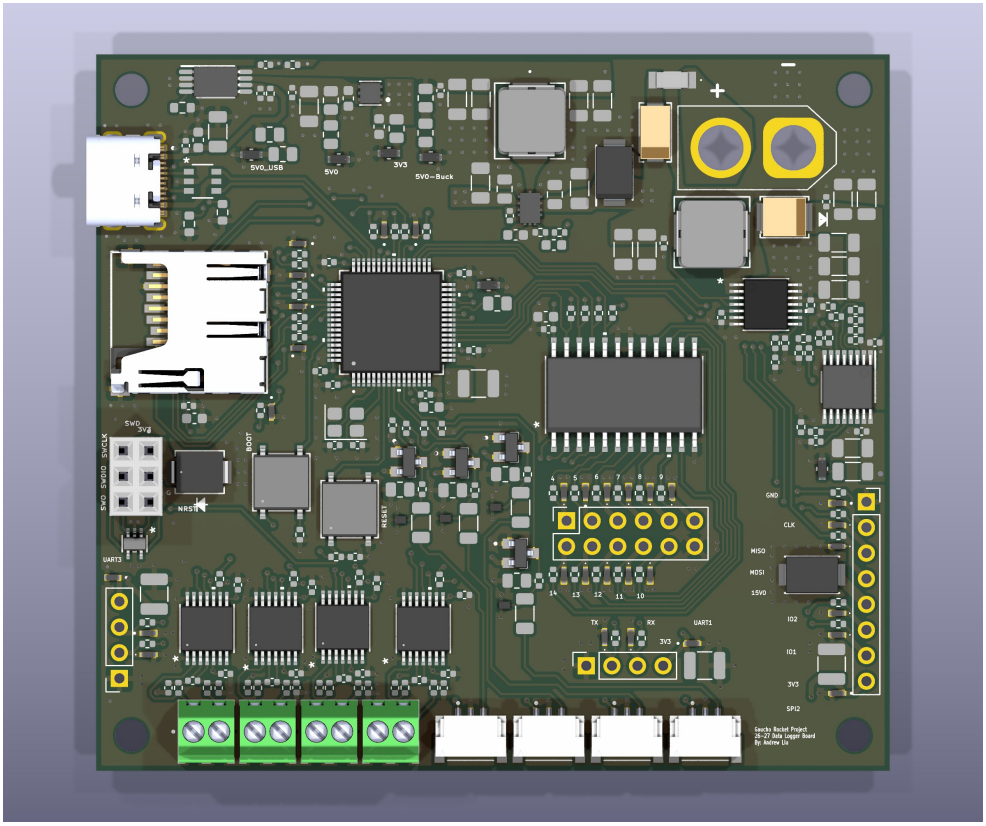


Figure 1: 3D render of the finished board (top side). The board is 78.7 mm × 71.0 mm, 4 layers, all parts on the top side.

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1 Introduction

This document records the design choices behind the 26–27 data logger board ("Hotfire Board"). The board logs up to four 4–20 mA pressure transducers and up to fifteen thermocouple amplifiers (four on board, eleven off board) to an SD card, using an STM32L496 microcontroller. It runs from a single 3S LiPo and can also be powered and programmed over USB-C.

The design went through one major revision. The first version ran from a flexible battery range (up to a 24 V pack) and used two LMR36006 bucks. The current version locks the input to a **3S 11.1 V LiPo only**, which allowed a much simpler power tree:

- a 5 V buck feeding a power mux with USB, then a low-noise LDO for a clean 3.3 V rail;
- a 15 V boost for the pressure transducers, followed by an eFuse that controls the turn-on ramp and limits current;
- a 4-to-16 chip-select decoder so many SPI devices can share one bus.

Every value in this write-up was taken from the KiCad project. Where the current schematic differs from the earlier draft write-up, the schematic is treated as correct and the difference is listed in Section 11.

1.1 Block diagram

Battery path: XT60 → 1 A fuse (F1) → reverse-polarity Schottky (CR1) → `VBAT` (SMBJ15A TVS, D24)
→ LMR36006 buck (U1) → `5V_BUCK` → TPS2113A mux (U4, IN1)
→ TPS55340 boost (U2) → `15V0_Boost` → TPS26600 eFuse (U12) → `15V0` (transducers, SMBJ18A D25)
USB path: USB-C (J6) → `5V_USB` → TPS2113A mux (U4, IN2)
Logic: TPS2113A OUT → `5V0` → LP5912 LDO (U5) → `3V3` (MCU, SD card, MAX31856s, decoder)
Data: STM32L496 (U3) → SDMMC (SD card), SPI2 + CD74HC154 (thermocouples, expansion),
ADC (4 pressure channels), USART1/USART3, USB FS, SWD

2 Power budget

The budget uses the measured maximum loads: 150 mA on the 3.3 V rail (MCU, SD card and thermocouple amplifiers) and 20 mA on the 15 V transducer rail. Converter efficiencies are assumed to be 85 %.

Stage	Load	Output power	From battery
15 V rail (boost, 85 %)	20 mA at 14.9 V	0.30 W	0.35 W
3.3 V rail (LDO from 5 V)	150 mA at 3.3 V	0.50 W	
LDO loss	$(5 - 3.3)\,\text{V} \times 150\,\text{mA}$	0.26 W	
5 V buck (85 %)	150 mA at 5 V	0.75 W	0.88 W
Total			**≈ 1.2 W**

Table 1: Power budget. 1.2 W is about 0.11 A from a 3S pack at 11.1 V (0.12 A at the 9.9 V cutoff). The 5 V and 15 V indicator LEDs add up to about 0.3 W more when lit.

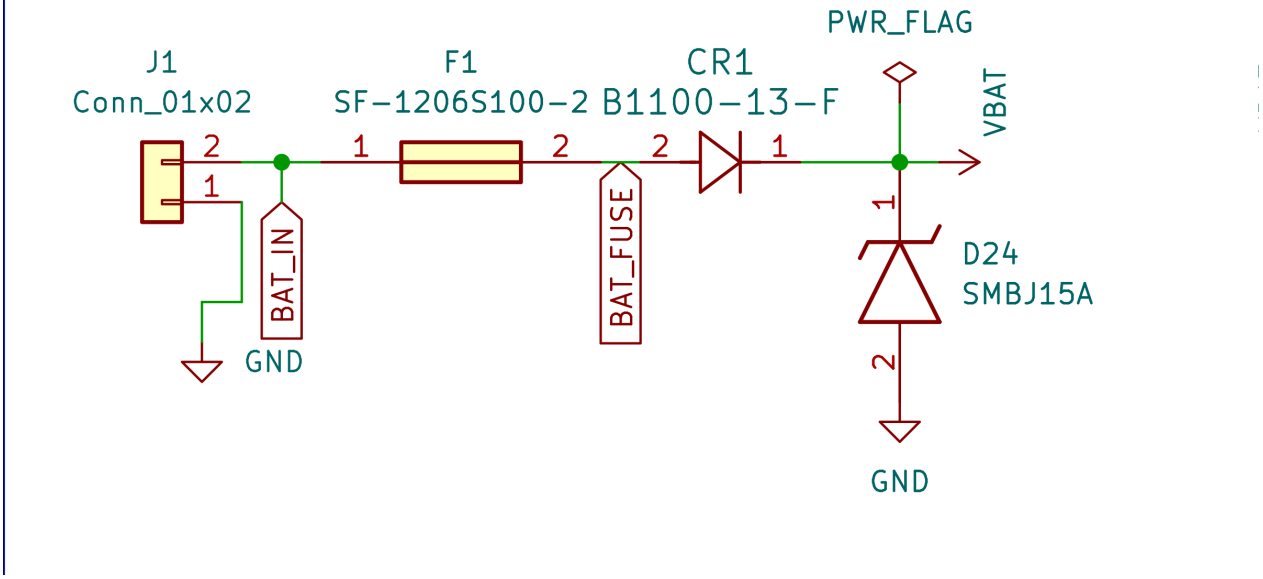
This is far inside every rating on the power path: the 1 A fuse and 1 A reverse-polarity diode, the LP5912's 500 mA, and the LMR36006's 0.6 A. The 15 V boost was sized for a 0.5 A design load (see Section 3.3), which leaves room for many more transducers on the expansion header.

3 Power path

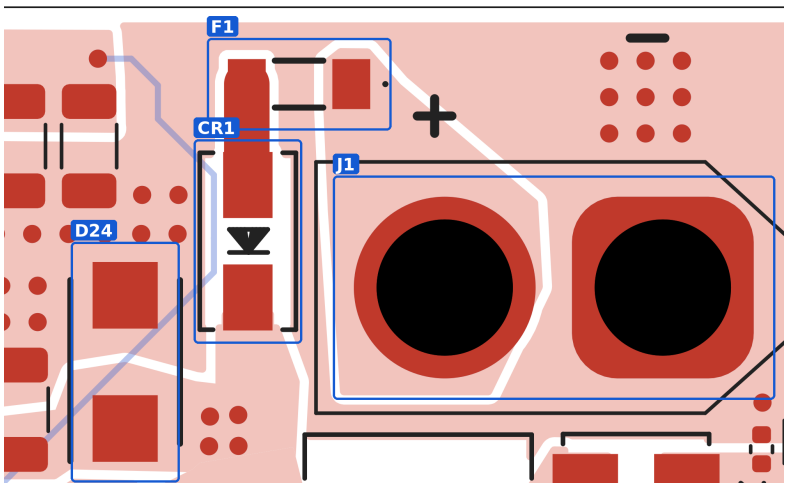
3.1 Battery input and protection

The battery is a 3S LiPo: 12.6 V full, 11.1 V nominal, and 9.9 V at a 3.3 V/cell cutoff. Locking to this one pack means every rail on the board can be built from a single boost and a single buck.

- **Connector (J1, Amass XT60PB-M.G.Y, LCSC C19268037):** vertical male PCB-mount XT60, mating with the female XT60 already on the battery lead, so no adapter is needed. On a rocket every adapter is another joint that can work loose. Tie the battery lead to the airframe for strain relief, since the XT60 is friction-fit only.
- **Fuse (F1, Bourns SF-1206S100-2):** 1 A fast-acting 1206 fuse in case of a short.
- **Reverse-polarity diode (CR1, B1100-13-F):** 100 V, 1 A Schottky in series, so a reversed battery can't damage anything.
- **TVS (D24, SMBJ15A):** 15 V standoff (above the 12.6 V full pack), about 24 V worst-case clamp. The clamp stays below the TPS55340's 34 V VIN absolute maximum and far below the LMR36006's 66 V.



(a) Schematic



(b) Layout (top copper red, bottom copper blue, poured copper light red; part outlined in blue)

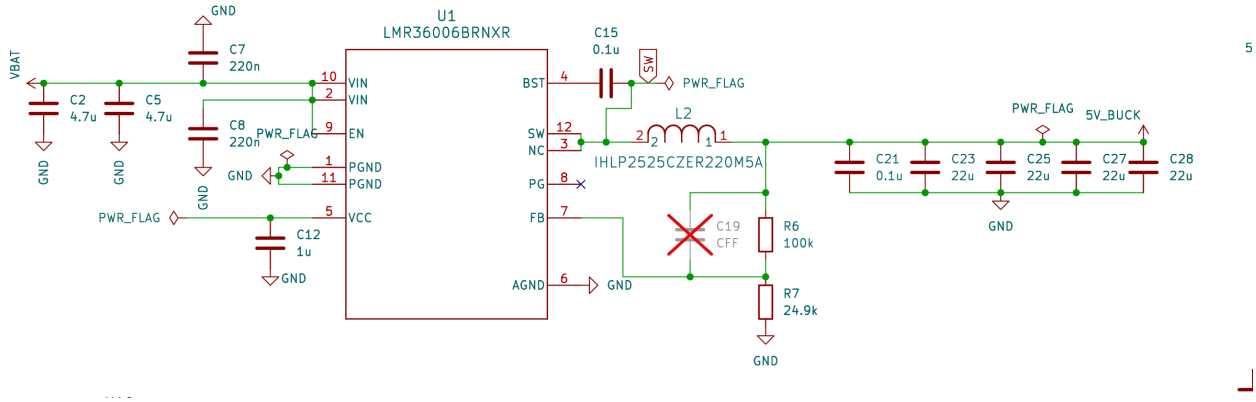
Figure 2: Battery input: XT60 (J1), fuse (F1), reverse-polarity Schottky (CR1) and TVS (D24).

3.2 5 V buck: LMR36006 (U1)

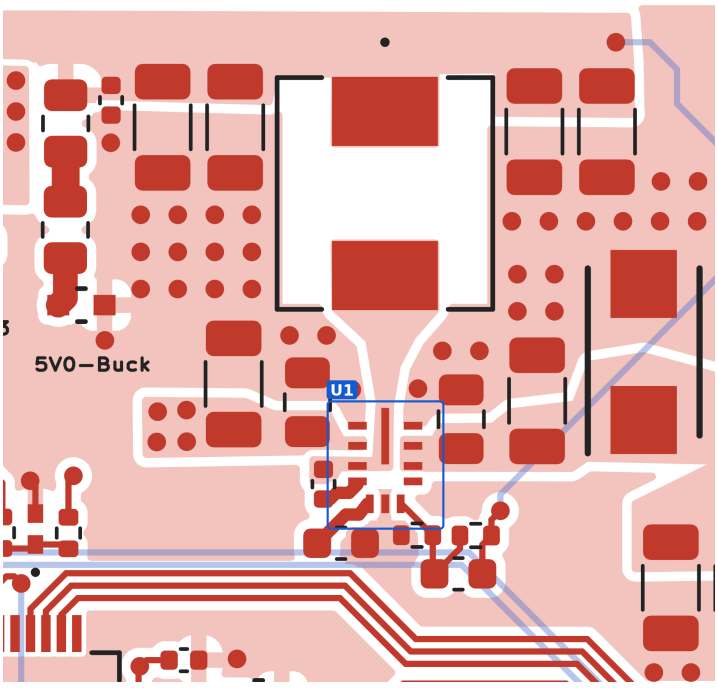
The LMR36006BRNXR is a 60 V, 0.6 A synchronous buck with internal FETs, so no external MOSFETs are needed. It was kept from the first version of the board. It is the 1 MHz variant. In the first version it made 3.3 V directly for the MCU; now it makes 5 V to match USB, so the mux compares like with like and a single low-noise LDO cleans up both sources.

Pin	Connection	Reason
VIN (2, 10)	`VBAT`	Battery after fuse and reverse diode
EN (9)	`VBAT`	Same voltage rating as VIN, so the buck runs whenever a battery
FB (7)	R6 100 kΩ / R7 24.9 kΩ	$V_{OUT} = 1\,\text{V} \times (1 + 100/24.9) = 5.02\,\text{V}$
BST (4)	C15 0.1 µF to SW	Bootstrap cap per datasheet
VCC (5)	C12 1 µF	Internal LDO decoupling
PG (8)	not connected	An LED (LED3) shows power instead
SW (3, 12)	L2 22 µH (IHLP2525CZER220M5A)	Switch node
PGND/AGND	`GND`	Analog pins return to AGND

- **Feedback:** with $R_{FBT} = 100\,\text{k}\Omega$ and $V_{REF} = 1\,\text{V}$, $R_{FBB} = 100\text{k}/(5/1 - 1) = 25\,\text{k}\Omega$, so $24.9\,\text{k}\Omega$ (the datasheet's own 5 V example). The 3.3 V version used $43.2\,\text{k}\Omega$.
- **Inductor:** the 3S input range puts L at about 13 µH to 17 µH for $K = 0.3$–0.4. The same 22 µH part as the boost is used to keep the BOM short; ripple is smaller and the saturation current is well above the 1.2 A high-side current limit.
- **Output caps:** 4×22 µF 1206 (C23, C25, C27, C28) plus 0.1 µF (C21). They are rated 10 V so DC bias doesn't eat most of the capacitance at 5 V. The mux and LDO caps also hang on this net; the total stays well under the datasheet's limit of about 10× the design value.
- **Input caps:** 2×4.7 µF 1206 (C2, C5) and 2×220 nF (C7, C8).
- **C_{FF} (C19):** footprint placed, not fitted.



(a) Schematic



(b) Layout (top copper red, bottom copper blue, poured copper light red; part outlined in blue)

Figure 3: LMR36006 5 V buck (U1).

3.3 15 V boost: TPS55340 (U2)

The 4–20 mA pressure transducers need a supply well above the battery, so a boost makes 14.9 V. Because the battery is always below 15 V, a plain boost works and no SEPIC or buck-boost is needed.

The TPS55340PWPR (HTSSOP-14) was chosen because it is cheap, has a 40 V/5 A internal switch, tolerates the SMBJ15A clamp, and has an adjustable frequency so it can be kept away from the buck's 1 MHz.

Pin	Connection	Reason
VIN (3), EN (4)	`VBAT`	Always on with a battery
SW (1, 2)	L1 22 µH → SS34 (U10)	Boost inductor and catch diode
SS (5)	C10 10 nF	≈3 ms soft start
SYNC (6)	`GND`	Frequency set by resistor only
FREQ (10)	R64 68.1 kΩ	≈693 kHz
FB (9)	R5 43.2 kΩ + R62 68.1 kΩ / R63 10 kΩ	14.91 V
COMP (8)	R2 590 Ω + C13 270 nF, C16 100 pF	Loop compensation
PGND, EPAD, AGND, NC	`GND`	

Switching frequency. Two free-running converters on the same input can beat at their difference frequency. With the LMR36006 at 1 MHz (harmonics 1, 2, 3 MHz), 700 kHz keeps every harmonic pair at least about 100 kHz apart (1.4, 2.1, 2.8 MHz), while 500 kHz, 600 kHz and 750 kHz would land exactly on a multiple. From the datasheet, $R_{FREQ}(\text{k}\Omega) = 57500 \times f_{SW}(\text{kHz})^{-1.03} = 67.5\,\text{k}\Omega$, so 68.1 kΩ (E96) gives about 693 kHz.

Output voltage. $V_{OUT} = 1.229\,\text{V} \times (1 + R_{SH}/R_{SL})$. To reuse resistors already on the BOM, $R_{SH} = 43.2\,\text{k}\Omega + 68.1\,\text{k}\Omega = 111.3\,\text{k}\Omega$ and $R_{SL} = 10\,\text{k}\Omega$:

$$V_{OUT} = 1.229 \times (1 + 111.3/10) = 14.91\,\text{V}.$$

Soft start. $t_{SS} = C_{SS} \times 1.8\,\text{V}/6\,\mu\text{A} = 3\,\text{ms}$ with the 10 nF part (CC0402KRX7R9BB103, already on the BOM). Note that a boost passes the battery through the inductor and diode before it starts switching, so the output jumps to about 12 V at plug-in; soft start only shapes the ramp from there to 14.9 V. The eFuse after the boost (Section 3.4) is what controls the ramp the transducers actually see.

Inductor (design point: 9.9 V in, 14.91 V/0.5 A out, $\eta = 0.85$).

$$D = \frac{V_{OUT} + V_D - V_{IN}}{V_{OUT} + V_D} = \frac{14.91 + 0.45 - 9.9}{15.36} = 0.355$$

$$I_{IN} = \frac{V_{OUT} I_{OUT}}{\eta V_{IN}} = 0.886\,\text{A}$$

$$L_{min} = \frac{V_{IN} D}{K\, I_{IN} f_{SW}} = \frac{9.9 \times 0.355}{0.3 \times 0.886 \times 693\text{k}} = 19.1\,\mu\text{H} \;\Rightarrow\; 22\,\mu\text{H}$$

$$\Delta I_L = 0.231\,\text{A}, \quad I_{L,rms} = 0.888\,\text{A}, \quad I_{L,peak} = 1.00\,\text{A}$$

The IHLP2525CZER220M5A (22 µH) has comfortable margin over the 1.0 A peak.

Output capacitors. Ripple (50 mV): $C_{OUT} \geq D I_{OUT}/(f_{SW} V_{ripple}) = 5.1\,\mu\text{F}$. Load transient (0.25 A step, 2 % droop, 10 kHz bandwidth): $C_{OUT} \geq 13.3\,\mu\text{F}$, which sets the requirement. Five 4.7 µF 1206 caps (C20, C22, C24, C72, C73) give about 15 µF effective after DC bias, plus C18 0.1 µF at the diode. With ceramics the ESR (about 2 mΩ) is far below the 0.14 Ω limit. RMS current is 0.37 A.

Input capacitors. $I_{CIN,rms} = \Delta I_L/\sqrt{12} = 67\,\text{mA}$. Two 4.7 µF (C3, C6, about 7 µF effective) plus C9 0.1 µF give about 13 mV of ripple.

Schottky diode (U10, SS34). 40 V, 3 A, SMA. Normal loss is about $0.5\,\mathrm{V} \times 0.5\,\mathrm{A} = 0.25\,\mathrm{W}$; even at a pessimistic $1.2\,\mathrm{A}/0.7\,\mathrm{V}$ it is 0.84 W, well within a 3 W part.

Loop compensation. Following the datasheet procedure with $G_{EA} = 440\,\mu\mathrm{S}$ (max), $R_{SENSE} = 15\,\mathrm{m}\Omega$ and 15 µF effective output capacitance:

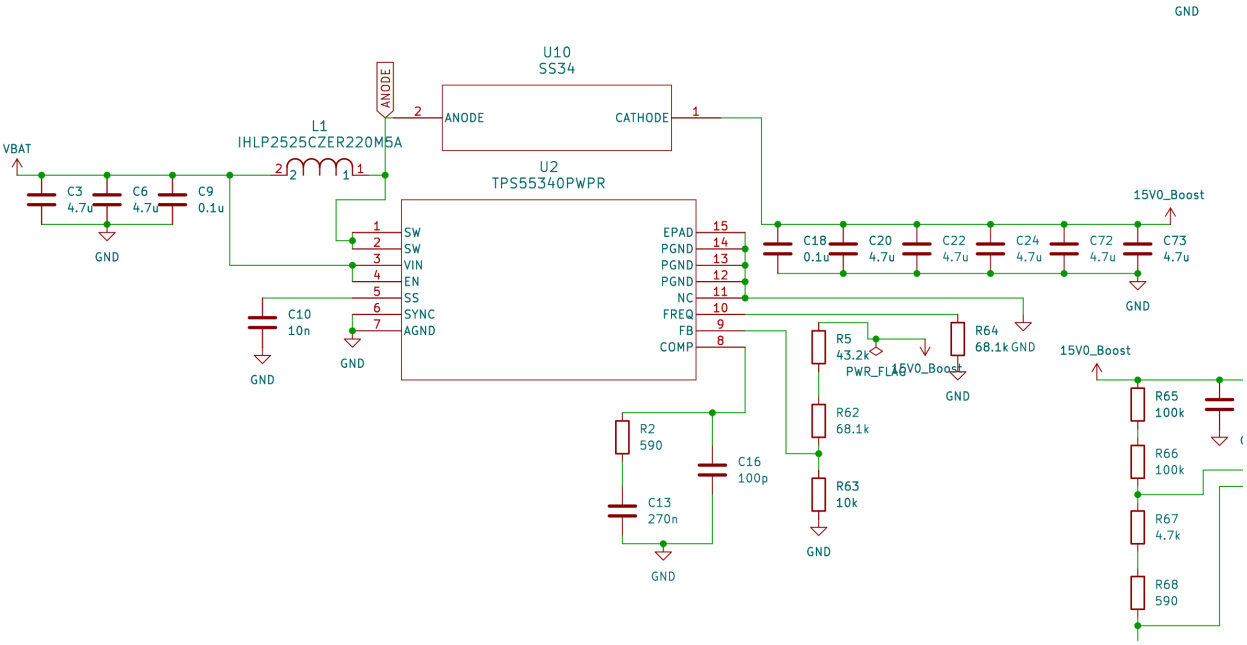
$$R_{OUT} = V_{OUT}/I_{OUT} = 29.8\,\Omega, \qquad f_{P,out} = \frac{2}{2\pi R_{OUT} C_{OUT}} = 712\,\mathrm{Hz},$$

$$f_{RHPZ} = \frac{R_{OUT}}{2\pi L}\left(\frac{V_{IN}}{V_{OUT}}\right)^2 = 95.1\,\mathrm{kHz}, \qquad f_{co} < \min(f_{SW}/5,\ f_{RHPZ}/3) = 31.7\,\mathrm{kHz}.$$

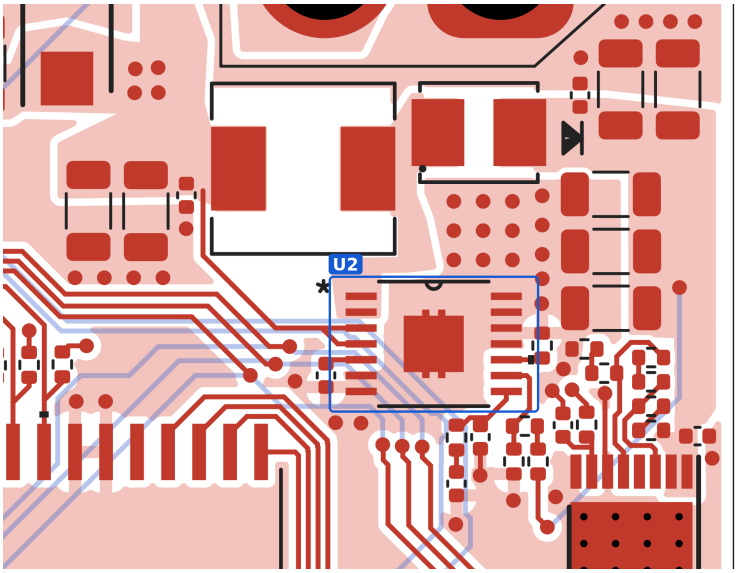
The crossover was set to 10 kHz for margin. The power-stage gain there is $K_{PS} = 47.1$ (33.5 dB), giving

$$R_3 = \frac{1}{G_{EA}\frac{R_{SL}}{R_{SH}+R_{SL}}K_{PS}} = 585\,\Omega \Rightarrow 590\,\Omega, \qquad C_4 = \frac{1}{2\pi R_3 (f_{co}/10)} = 270\,\mathrm{nF}, \qquad C_5 = \frac{R_{ESR}C_{OUT}}{R_3} = 51\,\mathrm{pF} \Rightarrow$$

100 pF was chosen for extra high-frequency filtering on COMP. On paper this gives about 80° of phase margin across 9.9 V to 12.6 V in and 0.1 A to 0.5 A out.



(a) Schematic



(b) Layout (top copper red, bottom copper blue, poured copper light red; part outlined in blue)

Figure 4: TPS55340 15 V boost (U2) with SS34 diode (U10) and 22 µH inductor (L1).

3.4 15 V eFuse / soft start: TPS26600 (U12)

Because the boost passes battery voltage straight through at plug-in, the transducers would see an uncontrolled step to about 12 V. The TPS26600PWPR eFuse after the boost fixes that: it holds the output off until the boost is actually regulating, then ramps it from 0 V at a controlled rate, and it also limits current and cuts off on overvoltage. It replaced an earlier plan of a TPS22810 load switch plus a PTC.

Pin	Connection	Reason
IN (1, 2)	`15V0_Boost`, C74 0.1 µF	Boost output
OUT (15, 16)	`15V0`, C76 0.1 µF + C77 4.7 µF	Transducer rail
UVLO (3), OVP (5)	R65+R66 / R67+R68 / R69+R70 divider	Turn-on and overvoltage thresholds
$\overline{\text{SHDN}}$ (7)	R74 100 kΩ to `3V3`, C78 0.1 µF	Enabled only once logic power is up
DVDT (12)	C75 0.1 µF	≈13 ms output ramp
ILIM (11)	R72 10 kΩ + R73 4.7 kΩ	≈0.8 A current limit
MODE (6)	`GND`	Current limit with auto-retry
RTN (8), GND, PAD	`GND`	No reverse protection needed after the boost
IMON, $\overline{\text{FLT}}$, NC	not connected	

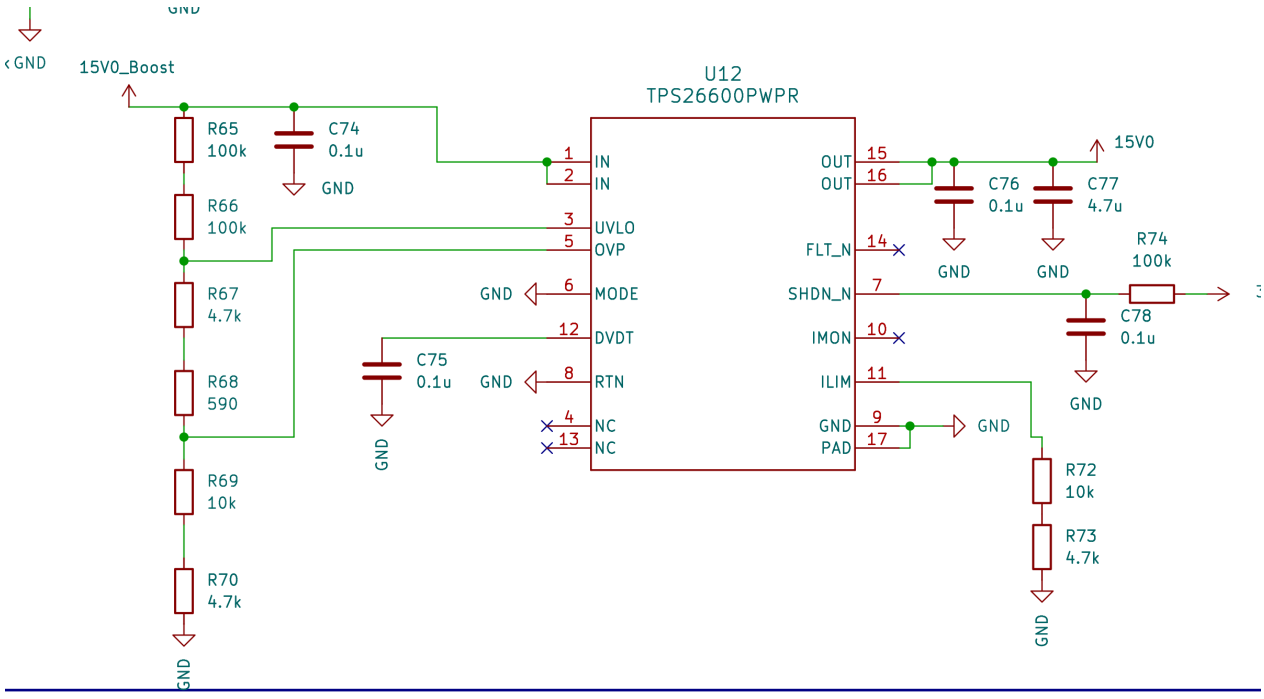
UVLO and OVP. A single three-resistor divider from IN sets both thresholds. To reuse BOM parts: $R_1 = 100\,\text{k}\Omega + 100\,\text{k}\Omega$, $R_2 = 4.7\,\text{k}\Omega + 590\,\Omega$, $R_3 = 10\,\text{k}\Omega + 4.7\,\text{k}\Omega$ (total 220 kΩ). With the datasheet's 1.19 V rising and 1.10 V falling thresholds:

$$V_{UV} = V_{th}\frac{R_1 + R_2 + R_3}{R_2 + R_3} = V_{th} \times 11.0 \qquad \Rightarrow \text{ on at } 13.1\,\text{V},\ \text{off at } 12.1\,\text{V}$$

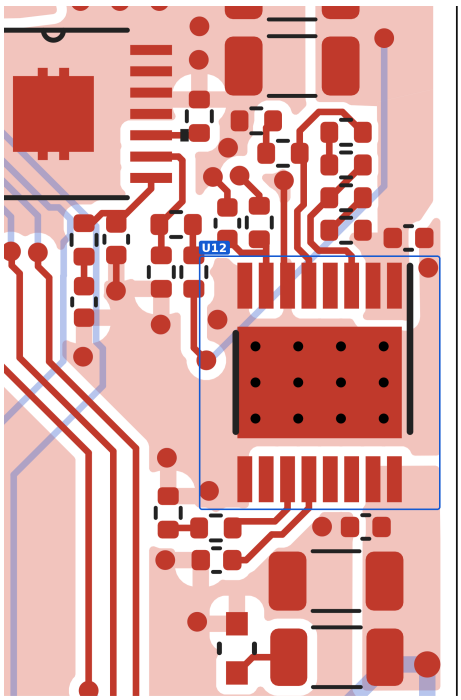
$$V_{OV} = V_{th}\frac{R_1 + R_2 + R_3}{R_3} = V_{th} \times 14.97 \qquad \Rightarrow \text{ trip at } 17.8\,\text{V},\ \text{recover at } 16.5\,\text{V}$$

The 13.1 V turn-on sits above the battery pass-through level (at most about 12.3 V), so the transducers only get power once the boost is regulating. The OVP trip is still inside the transducers' operating range.

Ramp, current limit and enable. 0.1 µF on DVDT gives a ramp of about 13 ms. $R_{ILIM} = 14.7\,\text{k}\Omega$ gives $I_{LIM} \approx 12/14.7 = 0.8\,\text{A}$. $\overline{\text{SHDN}}$ is pulled up to 3.3 V through 100 kΩ with a cap to ground, so the eFuse only enables once there is a logic supply and after a short RC delay. ($\overline{\text{SHDN}}$ is rated 4 V absolute max, so it is pulled to 3.3 V, not 5 V.)



(a) Schematic



(b) Layout (top copper red, bottom copper blue, poured copper light red; part outlined in blue)

Figure 5: TPS26600 eFuse on the 15 V transducer rail (U12).

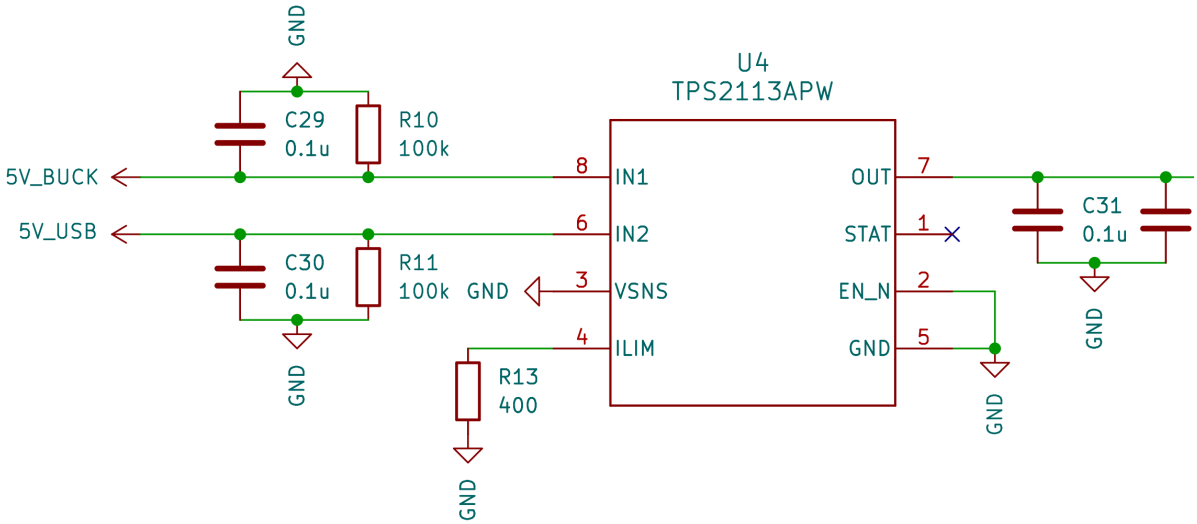
3.5 Power mux: TPS2113A (U4)

The board can run from the battery buck or from USB. The TPS2113APW switches between `5V_BUCK` (IN1) and `5V_USB` (IN2) and feeds `5V0`. Both inputs are 5 V, inside its 2.8 V to 5.5 V range, so the mux compares like with like. It also limits inrush, which smooths the switchover.

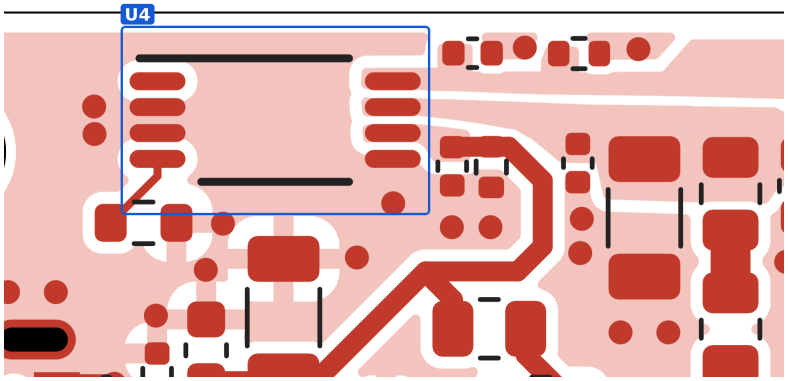
- **ILIM (4):** R13 400 Ω current-limit resistor.
- **$\overline{\text{EN}}$ (2) and VSNS (3):** tied to `GND`. With VSNS grounded the TPS2113A runs in auto-switching mode and powers `5V0` from whichever input is higher. Because the buck (5.02 V)

and USB are nearly equal, it may move between them as they drift; that is harmless (see below).

- **STAT (1):** not used.
- **Caps:** C29/C30 0.1 µF on each input, C31 0.1 µF + C32 22 µF on the output for a smooth switchover. At the 150 mA load, a switchover of about 10 µs dips the 22 µF by only $\Delta V = I\,t/C = 0.15 \times 10\,\mu\text{s}/22\,\mu\text{F} \approx 0.07\,\text{V}$. Even a much larger dip would not reach the 3.3 V rail, since the LDO has about 1.5 V of headroom.
- **Bleed resistors:** R10 and R11 (100 kΩ) discharge each input when its source is removed.



(a) Schematic



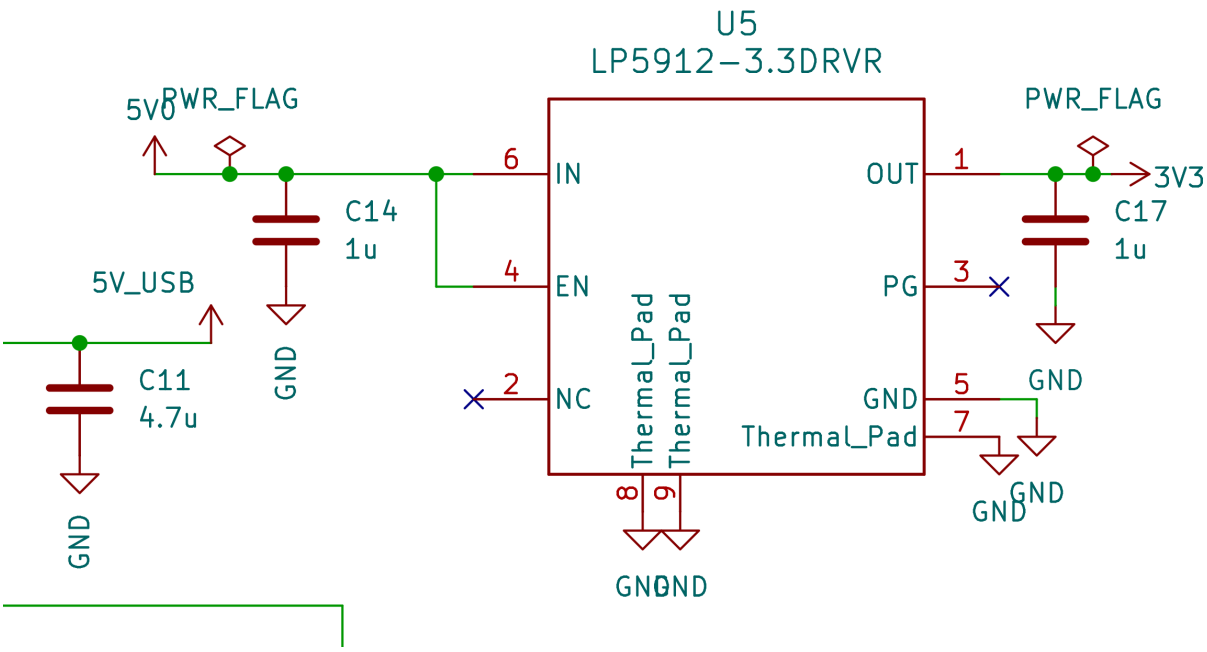
(b) Layout (top copper red, bottom copper blue, poured copper light red; part outlined in blue)

Figure 6: TPS2113A power mux (U4).

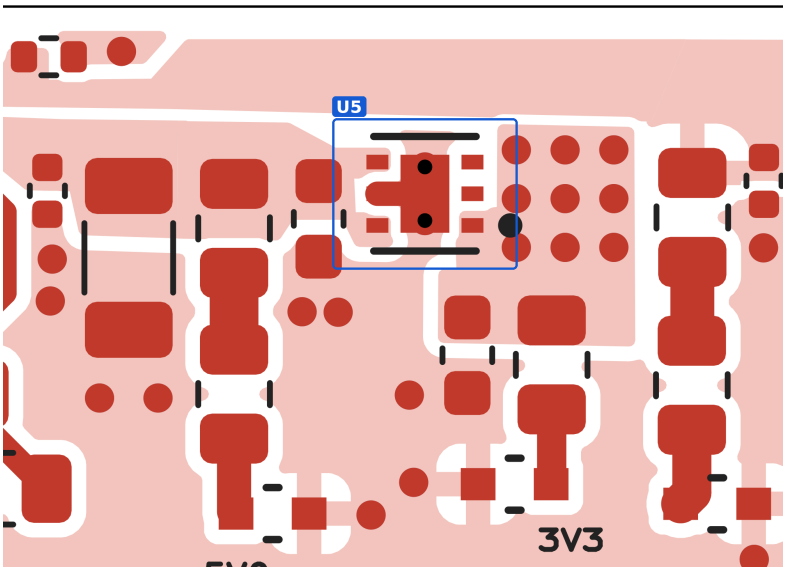
3.6 3.3 V LDO: LP5912-3.3 (U5)

A low-noise LDO after the mux gives the MCU, SD card and thermocouple amplifiers one clean 3.3 V rail whichever source is active, with high PSRR to the buck ripple and to noisy USB VBUS. The LP5912-3.3DRVR is rated 500 mA, over three times the 150 mA load. Going from 5 V to 3.3 V

leaves plenty of dropout headroom, and the LDO dissipates only about 0.26 W at the 150 mA load. EN is tied to IN; PG is not used. C14 (1 µF) is on the input and C17 (1 µF) on the output. A buck followed by an LDO has no loop interaction: each regulator runs its own loop and the LDO looks like an ordinary load to the buck.



(a) Schematic



(b) Layout (top copper red, bottom copper blue, poured copper light red; part outlined in blue)

Figure 7: LP5912 3.3 V LDO (U5).

3.7 Power indicator LEDs

All LEDs are LTST-C190KRKT (red, $V_F \approx 2$ V), sized for about 10 mA:

LED	Rail / signal	Resistor	Current
LED1	`3V3`	R1 130 Ω	10 mA
LED3	`5V_BUCK`	R71 + R14 (260 Ω)	11.5 mA
LED4	`5V0` (mux out)	R79 + R15 (260 Ω)	11.5 mA
LED6	`5V_USB`	R91 + R92 (260 Ω)	11.5 mA
LED2	`15V0`	R8 1.3 kΩ (1206)	10 mA, 0.13 W
LED5	STATUS (PB1)	R61 130 Ω	10 mA

4 Microcontroller: STM32L496RET6 (U3)

The STM32L496 (LQFP-64) was kept from the first board. It has enough SPI buses, an SDMMC peripheral for 4-bit SD, a 12-bit ADC, USB FS with DFU bootloader, and low power.

4.1 Supporting circuitry

- **Power and decoupling** (following ST AN4555): every VDD pin has 0.1 µF, with a 10 µF bulk cap (C35); VDDA/VREF+ has 1 µF + 10 nF (C39, C40). VDDUSB and VBAT are tied to 3.3 V (no backup battery).
- **HSE crystal (Y1):** 16 MHz 3225 crystal with 10 pF load caps (C33, C34) on PH0/PH1, for an accurate PLL clock for USB. The LSE pins (PC14/PC15) are unused.
- **Reset (SW1):** pushbutton to GND on NRST, with a 10 kΩ pull-up (R9) and 0.1 µF (C26).
- **Boot (SW2):** pushbutton to 3.3 V on BOOT0 (pin 60) with a 10 kΩ pull-down (R12). Holding BOOT while pressing RESET enters the ROM bootloader, so the board can be flashed over USB (DFU) without a debugger.
- **SWD (J2, Samtec BCS-103 2×3):** SWDIO, SWCLK, SWO and NRST for step-through debugging. CR5 (ESDALC6V1W5) protects the four signal lines and CR4 (SMLVT3V3) protects the 3.3 V pin.
- **USB-C (J6, GCT USB4105-GF-A):** 5.1 kΩ pull-downs on CC1/CC2 (R3, R4) mark the board as a sink. A USBLC6-2SC6 (CR3) protects D+/D− and VBUS. The D+/D− pair is routed with 0.27 mm traces for impedance control.
- **Status LED** on PB1 (LED5).

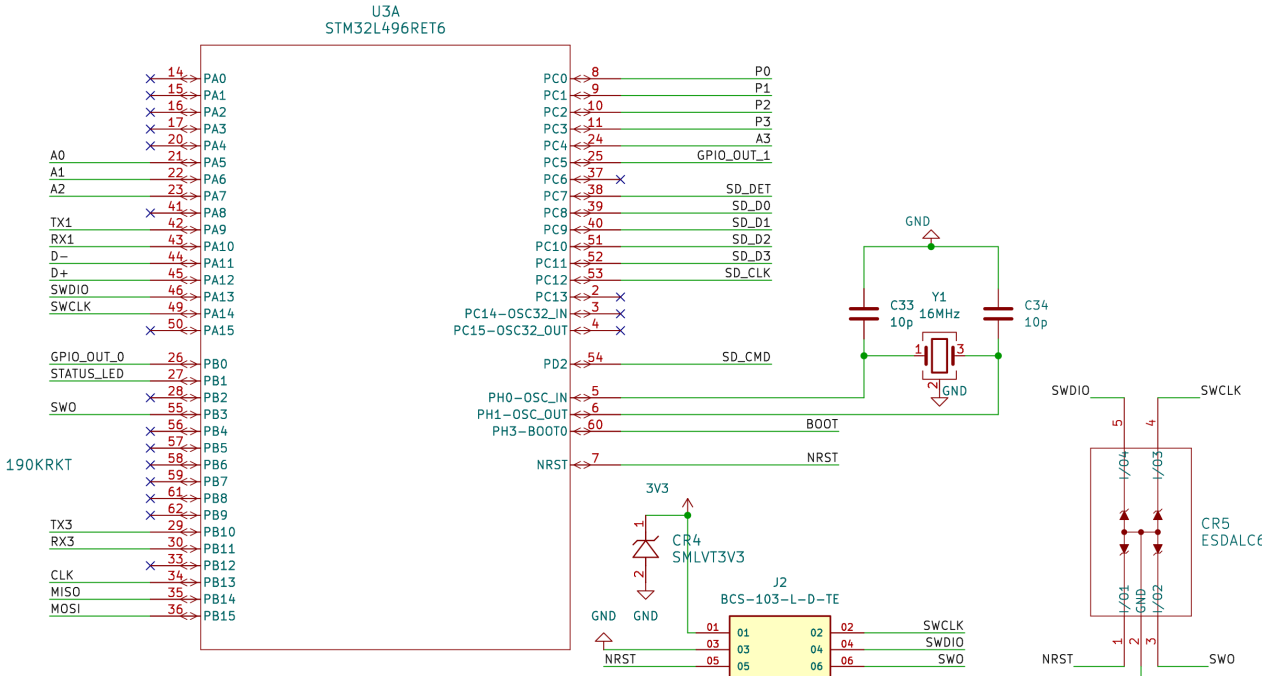
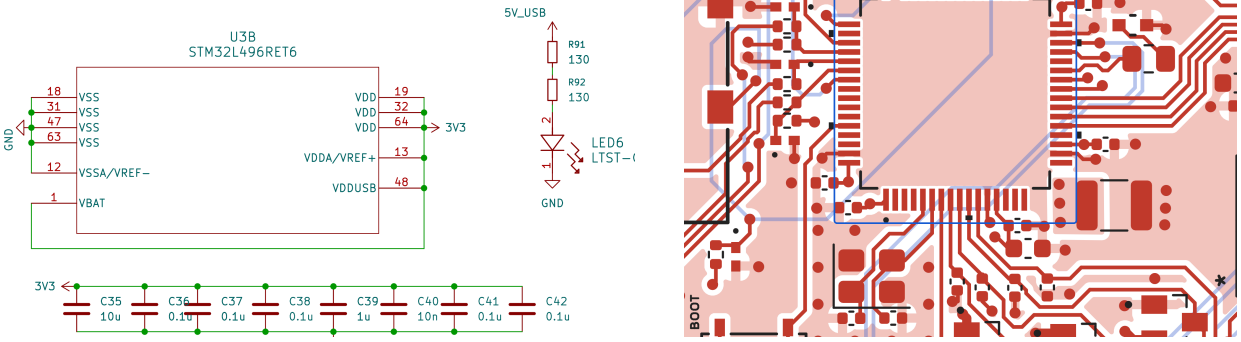


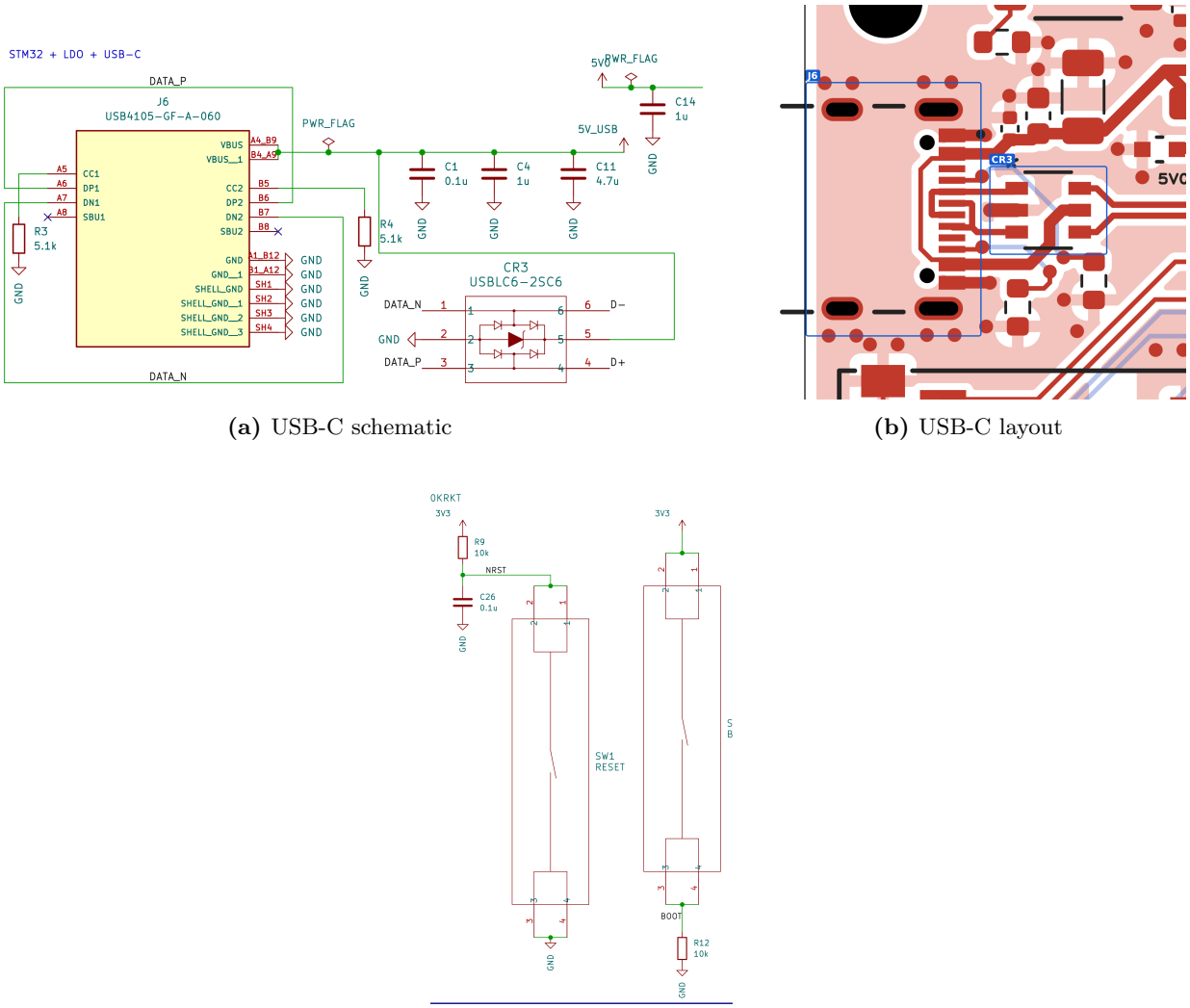
Figure 8: STM32L496 signal pins, HSE crystal and SWD header.



(a) Power pins and decoupling

(b) Layout around U3

Figure 9: STM32L496 power and layout.



(a) USB-C schematic

(b) USB-C layout

(c) Reset and boot buttons

Figure 10: USB-C connector and reset/boot circuitry.

4.2 Pin assignments

Table 2: STM32L496 (LQFP-64) pin assignments, taken from the netlist.

Pin	Port	Net	Function / connects to
5	PH0	OSC_IN	HSE 16 MHz crystal (Y1)
6	PH1	OSC_OUT	HSE crystal (Y1)
7	NRST	NRST	Reset button SW1, SWD J2-5
8	PC0	P0	ADC1_IN1, pressure channel J5
9	PC1	P1	ADC1_IN2, pressure channel J9
10	PC2	P2	ADC1_IN3, pressure channel J10
11	PC3	P3	ADC1_IN4, pressure channel J7
21	PA5	A0	Decoder address A0 (U11-23)
22	PA6	A1	Decoder address A1 (U11-22)
23	PA7	A2	Decoder address A2 (U11-21)
24	PC4	A3	Decoder address A3 (U11-20)
25	PC5	GPIO_OUT_1	Expansion header J15-7
26	PB0	GPIO_OUT_0	Expansion header J15-6
27	PB1	STATUS_LED	LED5

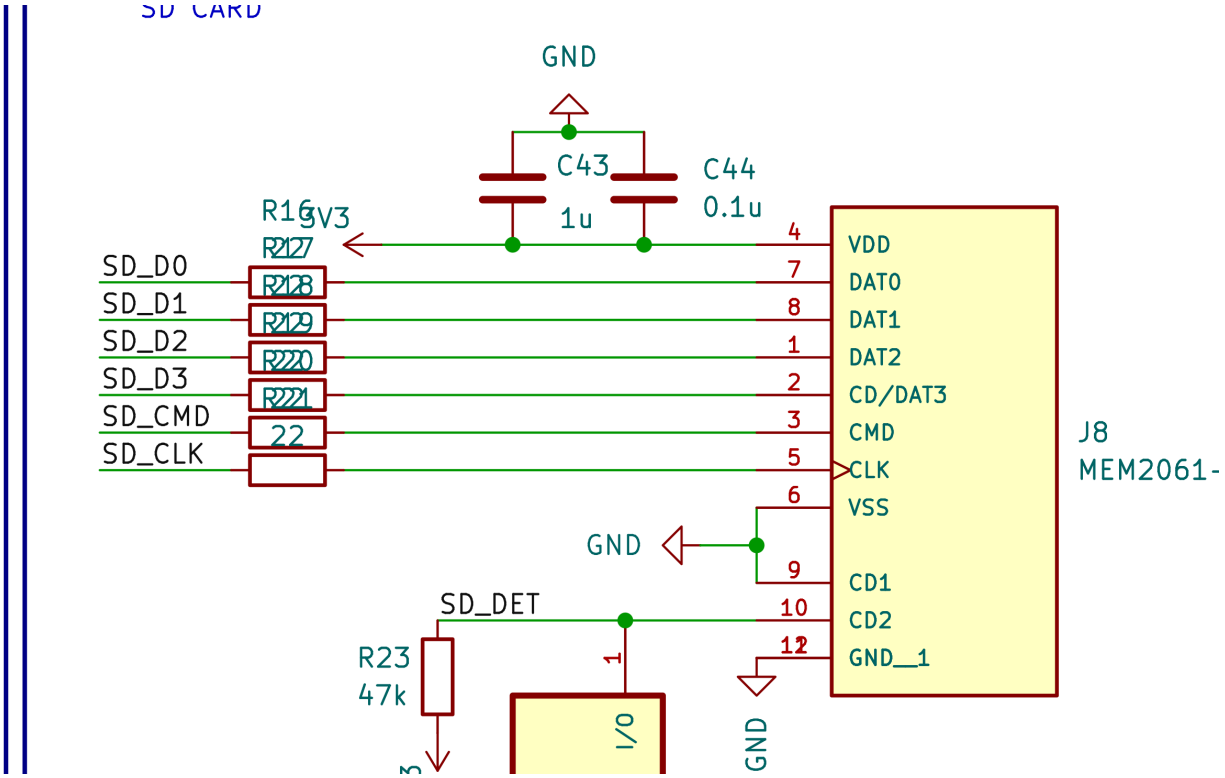
Pin	Port	Net	Function / connects to
29	PB10	TX3	USART3_TX, header J4-3
30	PB11	RX3	USART3_RX, header J4-2
34	PB13	CLK	SPI2_SCK: MAX31856s, J15-2
35	PB14	MISO	SPI2_MISO: MAX31856s, J15-3
36	PB15	MOSI	SPI2_MOSI: MAX31856s, J15-4
38	PC7	SD_DET	SD card detect
39	PC8	SD_D0	SDMMC1_D0
40	PC9	SD_D1	SDMMC1_D1
42	PA9	TX1	USART1_TX, header J3-2
43	PA10	RX1	USART1_RX, header J3-3
44	PA11	D−	USB_OTG_FS_DM
45	PA12	D+	USB_OTG_FS_DP
46	PA13	SWDIO	SWD J2-4
49	PA14	SWCLK	SWD J2-2
51	PC10	SD_D2	SDMMC1_D2
52	PC11	SD_D3	SDMMC1_D3
53	PC12	SD_CLK	SDMMC1_CK
54	PD2	SD_CMD	SDMMC1_CMD
55	PB3	SWO	SWD J2-6
60	PH3-BOOT0	BOOT	Boot button SW2
1	VBAT	3V3	Tied to 3.3 V
13 / 12	VDDA / VSSA	3V3 / GND	ADC supply and reference
19, 32, 64	VDD	3V3	
48	VDDUSB	3V3	
18, 31, 47, 63	VSS	GND	
Unused: PC13, PC14, PC15, PA0–PA4, PB2, PB12, PC6, PA8, PA15, PB4–PB9.			

Note: SPI2 was chosen for the thermocouple bus. The decoder address lines use PA5–PA7, which are also SPI1's pins, so SPI1 is not available as a second bus in this revision.

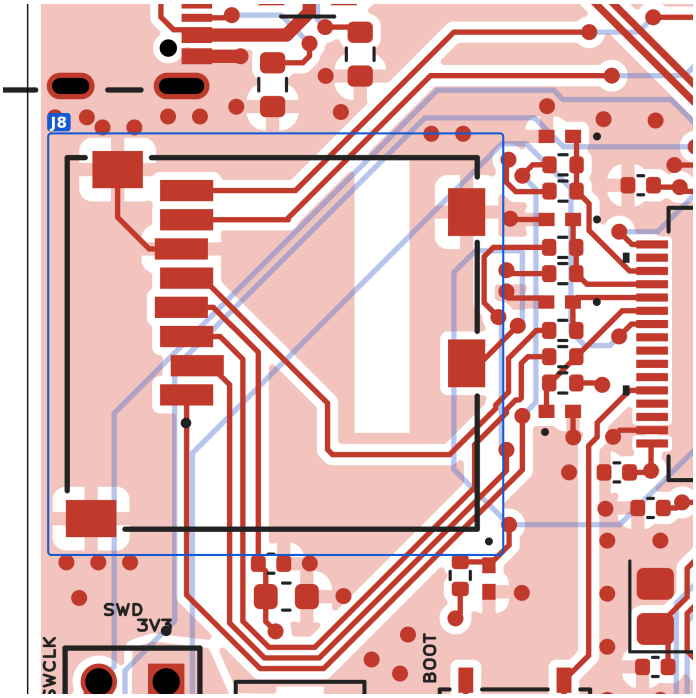
5 SD card

Data is logged to a microSD card (J8, GCT MEM2061) through the **SDMMC1** peripheral in 4-bit mode, which is much faster than SPI mode. An earlier plan used SPI3, but the SDMMC pins overlap with it, so SPI3 is not available.

- 22 Ω series resistors (R16–R21) on every data, CMD and CLK line to damp ringing.
- 47 kΩ pull-ups (R22, R24, R27, R32, R33) on CMD and D0–D3, as the SD spec requires. CLK has no pull-up.
- Card detect on PC7 with a 47 kΩ pull-up (R23).
- TPD1E05U06DPYR ESD diodes (D1–D5, D8, D9) on every line.
- Decoupling: C43 1 µF and C44 0.1 µF at the socket.



(a) Schematic



(b) Layout (top copper red, bottom copper blue, poured copper light red; part outlined in blue)

Figure 11: microSD socket (J8) with series resistors and pull-ups.

6 Thermocouple amplifiers: MAX31856 (U6–U9)

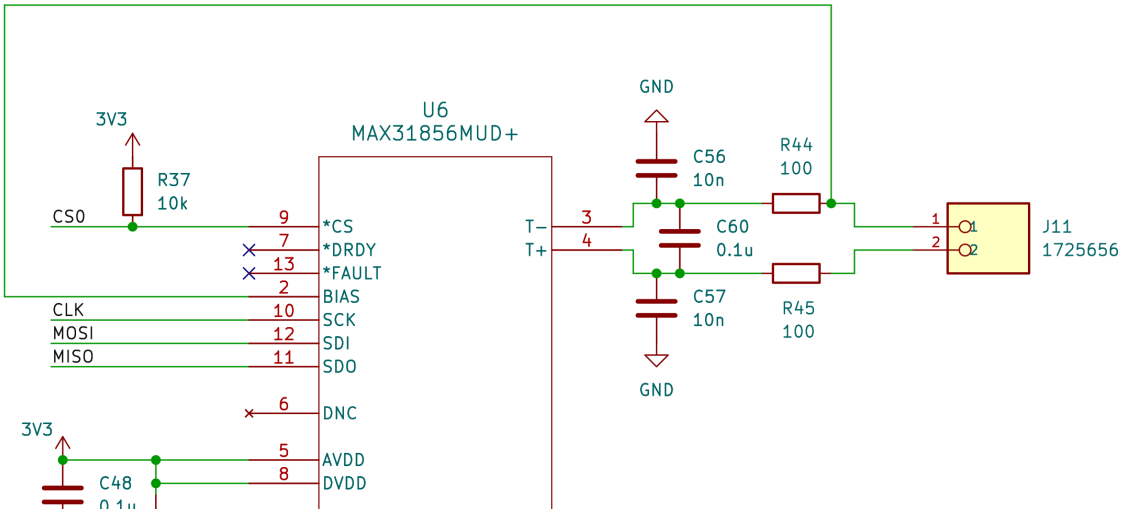
Four MAX31856MUD+ thermocouple-to-digital converters sit on SPI2. Each has its own Phoenix 1725656 screw terminal (J11–J14) for the thermocouple.

- **SPI:** SCK, SDI and SDO are shared on SPI2 (PB13/PB15/PB14). Each chip's $\overline{\text{CS}}$ comes from the decoder (Y0–Y3) and has a 10 kΩ pull-up (R37–R40), so every chip stays deselected

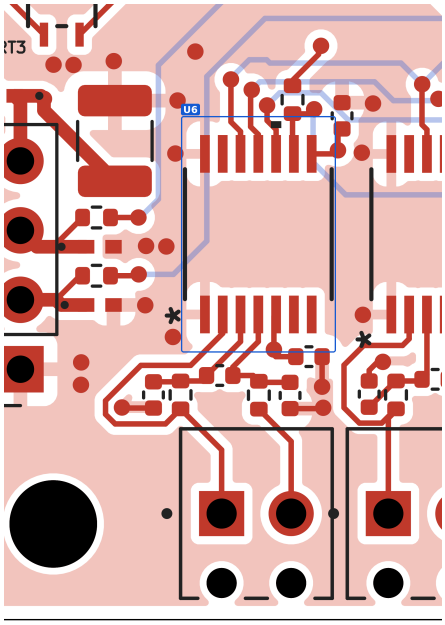
during boot.

- **Input filter:** 100 Ω in series with each thermocouple lead, 0.1 µF differential and 10 nF from each input to ground. Terminal pin 1 connects to BIAS and, through its 100 Ω, to T−; pin 2 goes through 100 Ω to T+.
- **$\overline{\text{DRDY}}$ and $\overline{\text{FAULT}}$:** left open. Conversion status and faults are read over SPI, which saves eight MCU pins.
- **Supply:** AVDD and DVDD on 3.3 V, each with 0.1 µF.

Amplifier	Terminal	Decoder output	CS pull-up	Address (A3..A0)
U6	J11	Y0 (CS0)	R37	0000
U7	J12	Y1 (CS1)	R38	0001
U8	J13	Y2 (CS2)	R39	0010
U9	J14	Y3 (CS3)	R40	0011

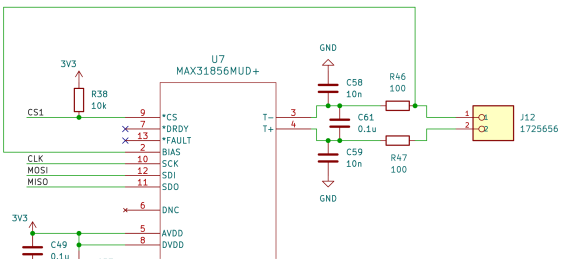


(a) Schematic

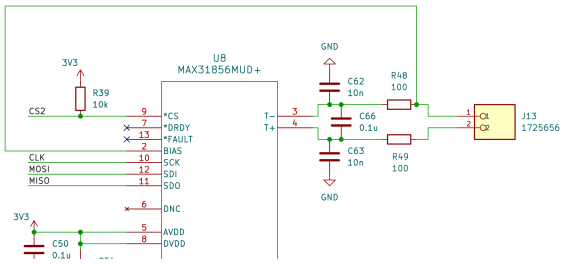


(b) Layout (top copper red, bottom copper blue, poured copper light red; part outlined in blue)

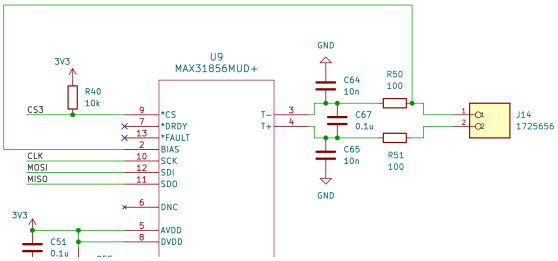
Figure 12: MAX31856 thermocouple amplifier, channel 1 (U6). Channels 2–4 (U7–U9) are identical.



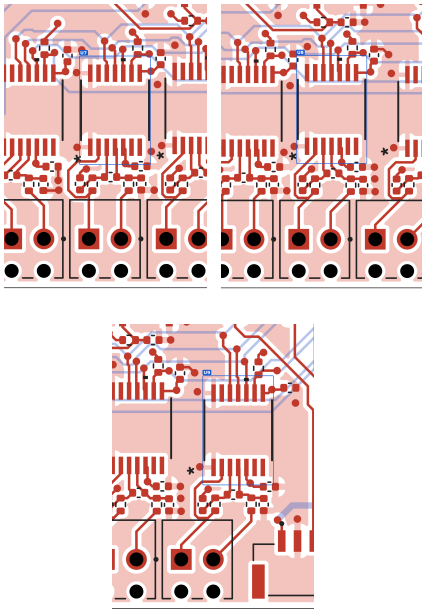
(a) U7 schematic



(b) U8 schematic



(c) U9 schematic



(d) U7, U8, U9 layout

Figure 13: MAX31856 channels 2–4.

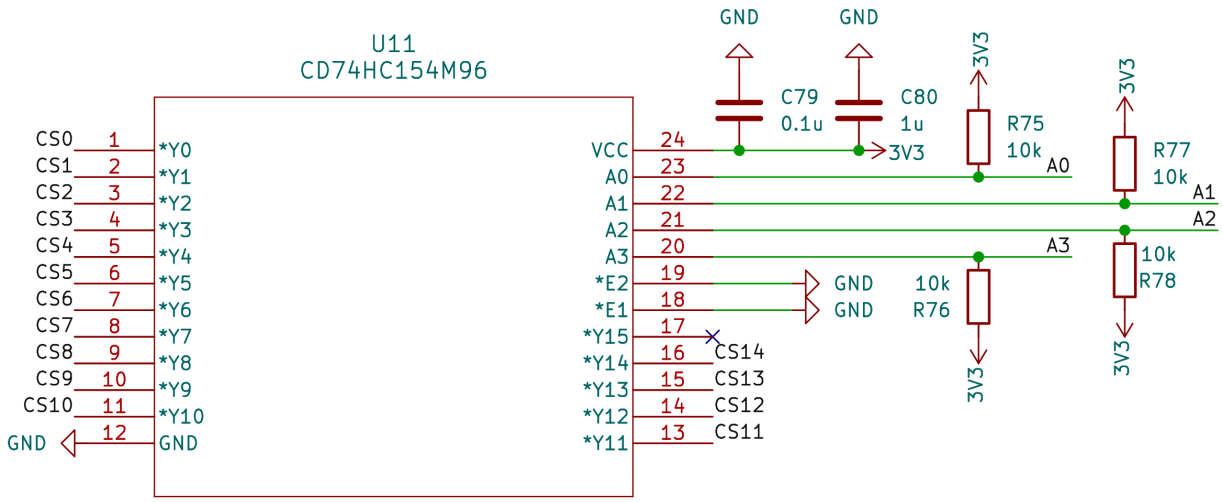
7 SPI chip-select decoder: CD74HC154 (U11)

Giving every SPI device its own chip-select pin would use 16 MCU pins. A 4-to-16 decoder (CD74HC154M96) needs only four address lines, and by design only one output can be low at a time, so two devices can never be selected together. Speed isn't a concern: a MAX31856 only converts about every 100 ms.

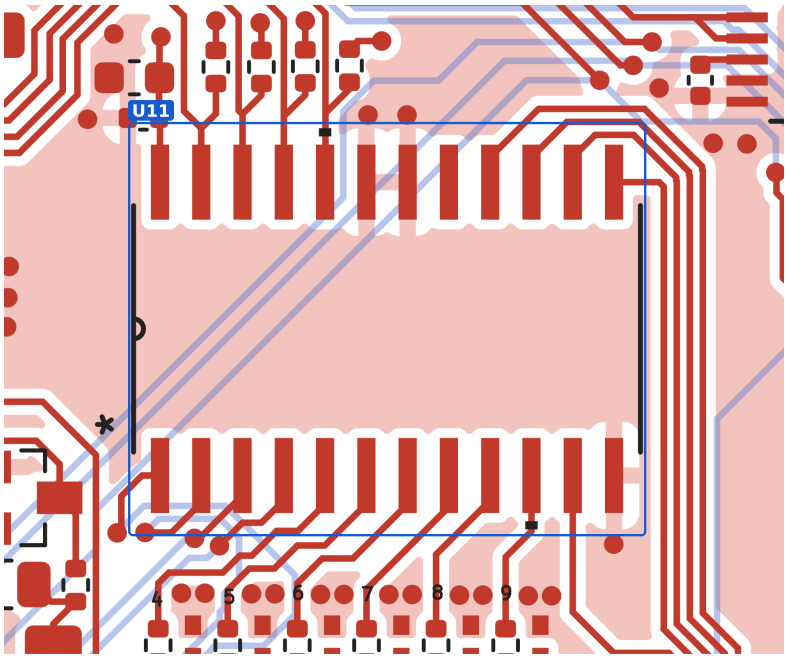
- **Address A0–A3:** PA5, PA6, PA7, PC4, each with a 10 kΩ pull-up to 3.3 V (R75–R78).
- **Enables $\overline{\text{E1}}$, $\overline{\text{E2}}$:** both tied to GND, so the decoder is always enabled.
- **Idle address:** Y15 is left unconnected and used as the "nothing selected" address. The pull-ups make 1111 the default, so no device is selected while the MCU boots or is in reset.
- **Outputs:** Y0–Y3 go to the on-board MAX31856s; Y4–Y14 go through 100 Ω resistors (R80–R90) and TPD1E05U06DPYR ESD diodes (D26–D36) to the 2×6 header J16. That gives **15 chip selects** in total.
- **Supply:** VCC on 3.3 V with C79 0.1 µF and C80 1 µF.

Firmware sequence. Start from the idle address 1111. Write the target address to A3..A0 in a single GPIO write (one `BSRR` write per port, so the address lines change together), run the SPI transfer, then write 1111 again. Returning to 1111 gives each MAX31856 the $\overline{\text{CS}}$ rising edge it needs to end a transaction. Because A3 is on port C and A0–A2 are on port A, update port A first when

leaving idle and port C first when returning, so the address never passes through another device's code.



(a) Schematic



(b) Layout (top copper red, bottom copper blue, poured copper light red; part outlined in blue)

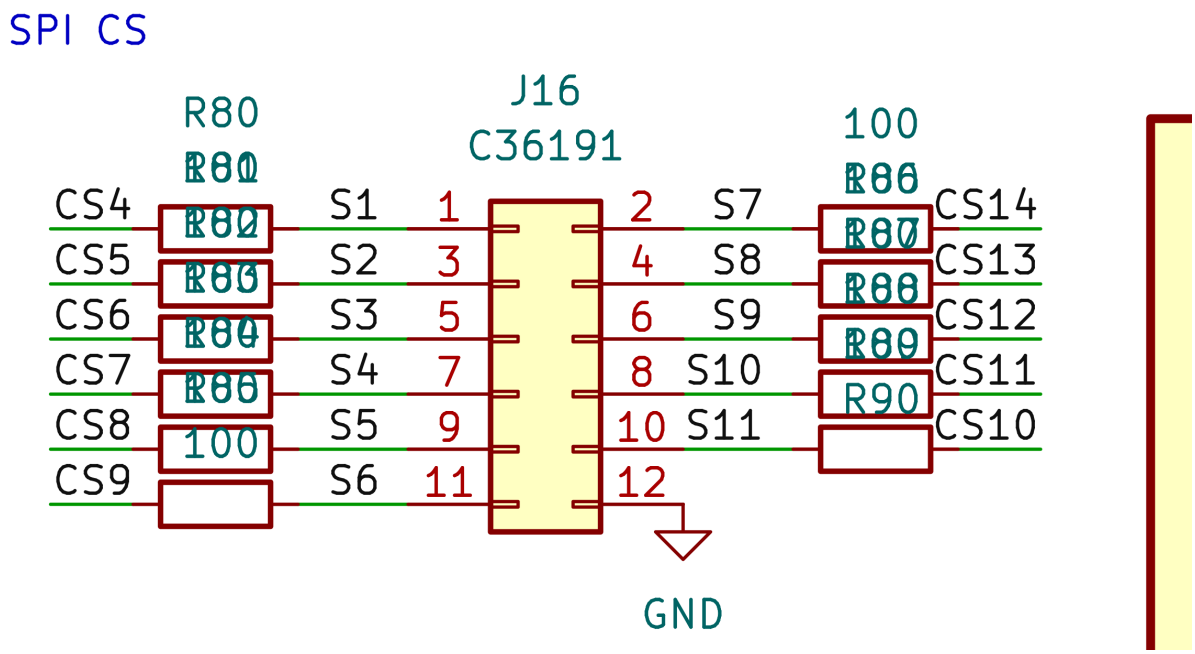
Figure 14: CD74HC154 4-to-16 chip-select decoder (U11).

7.1 Chip-select header (J16)

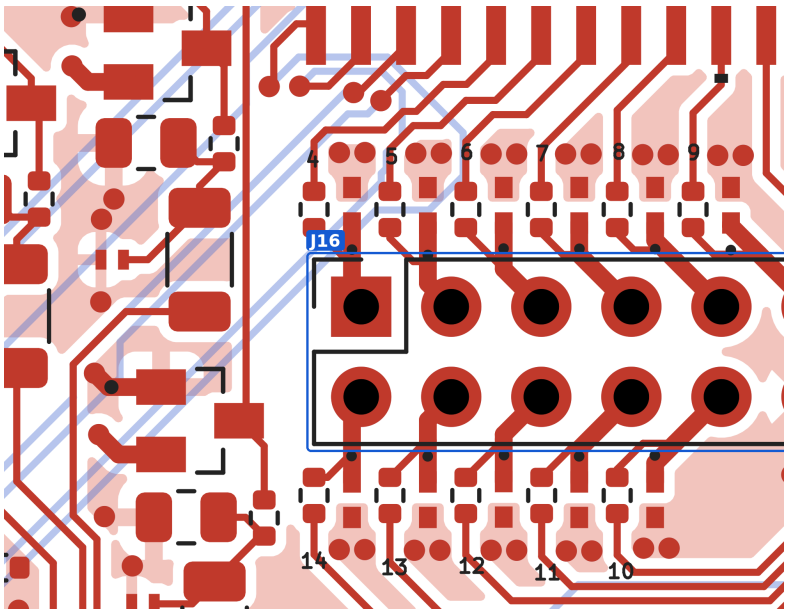
J16 is a 2×6, 2.54 mm socket (LCSC C36191) carrying the eleven off-board chip selects and ground.

J16 pin	Signal	Decoder	J16 pin	Signal	Decoder
1	S1	Y4 (CS4)	2	S7	Y14 (CS14)
3	S2	Y5 (CS5)	4	S8	Y13 (CS13)
5	S3	Y6 (CS6)	6	S9	Y12 (CS12)
7	S4	Y7 (CS7)	8	S10	Y11 (CS11)
9	S5	Y8 (CS8)	10	S11	Y10 (CS10)
11	S6	Y9 (CS9)	12	GND	

Table 3: J16 pinout. Address = decoder output number in binary, e.g. S1 = Y4 = 0100.



(a) Schematic



(b) Layout (top copper red, bottom copper blue, poured copper light red; part outlined in blue)

Figure 15: Chip-select header J16 with series resistors.

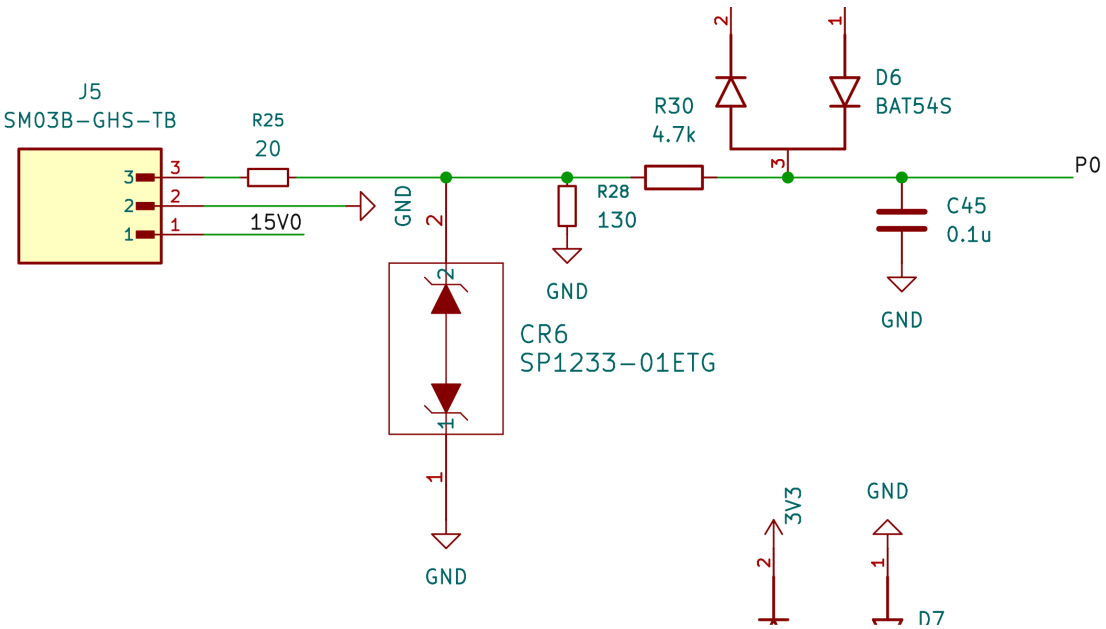
8 Pressure transducer inputs

The four 4–20 mA transducers connect through JST SM03B-GHS-TB 3-pin connectors (J5, J7, J9, J10): pin 1 is 15 V from the eFuse, pin 2 GND, pin 3 the loop current return. The first board used diode clamps to the rails and relied on the dev boards' own protection. This board is bare, so each channel gets a fuller protection chain:

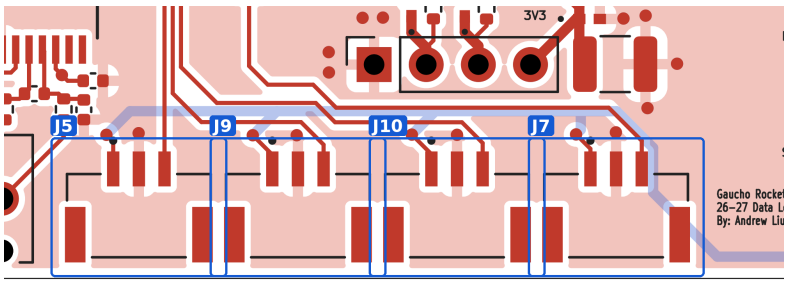
1. **20 Ω series resistor** (1206 anti-surge, RT1206FRE0720RL) limits current into the TVS.
2. **SP1233-01ETG TVS** to GND at the connector.
3. **130 Ω shunt** (ERJP06F1300V, 0805 anti-surge) converts 4–20 mA to 0.52 V to 2.6 V, inside the 3.3 V ADC range.
4. **4.7 kΩ + 0.1 µF RC** into the ADC pin ($f_c \approx 340$ Hz), which also limits clamp current.
5. **BAT54S dual Schottky** clamping the ADC pin between GND and 3.3 V.

Connector	Series R	TVS	Shunt	Filter R	Filter C	Clamp	ADC pin
J5	R25	CR6	R28	R30	C45	D6	PC0 (P0)
J9	R34	CR8	R35	R36	C47	D10	PC1 (P1)
J10	R41	CR9	R42	R43	C68	D11	PC2 (P2)
J7	R26	CR7	R29	R31	C46	D7	PC3 (P3)

With the transducers on 14.9 V and up to 2.6 V across the shunt plus 0.4 V across the series resistor, each transducer still has about 11.9 V, enough for typical 4–20 mA transmitters. Plug the transducers in before the battery.



(a) Schematic



(b) Layout (top copper red, bottom copper blue, poured copper light red; part outlined in blue)

Figure 16: Pressure input channel 1 (J5 → PC0). The other three channels are identical; all four connectors are outlined in the layout.

9 Breakout headers

Every signal that leaves the board passes through a 100 Ω series resistor and a TPD1E05U06DPYR ESD diode right at the pin, and these traces are 0.5 mm wide.

Header	Purpose	Pinout
J3 (1×4)	USART1, MCU-to-MCU or PC	1 GND, 2 TX1 (PA9), 3 RX1 (PA10), 4 3V3
J4 (1×4)	USART3, LoRa or other radio	1 GND, 2 RX3 (PB11), 3 TX3 (PB10), 4 3V3
J15 (1×8)	SPI2 expansion (future sensors)	1 GND, 2 CLK, 3 MISO, 4 MOSI, 5 15V0, 6 GPIO_OUT_0 (PB0), 7 GPIO_OUT_1 (PC5), 8 3V3
J2 (2×3)	SWD debug	1 3V3, 2 SWCLK, 3 GND, 4 SWDIO, 5 NRST, 6 SWO
J16 (2×6)	Chip selects	see Table above

- The 15 V pin on J15 comes from after the eFuse and has an **SMBJ18A** TVS (D25), whose breakdown is above the eFuse's 17.8 V OVP trip.
- Each header's 3.3 V pin has a 10 µF cap (C69, C70, C71).
- Future pressure transducers are meant to go on an expansion board with an SPI ADC (e.g. MCP3208 or ADS8344) over J15 plus a decoder chip select, rather than more ADC inputs here.

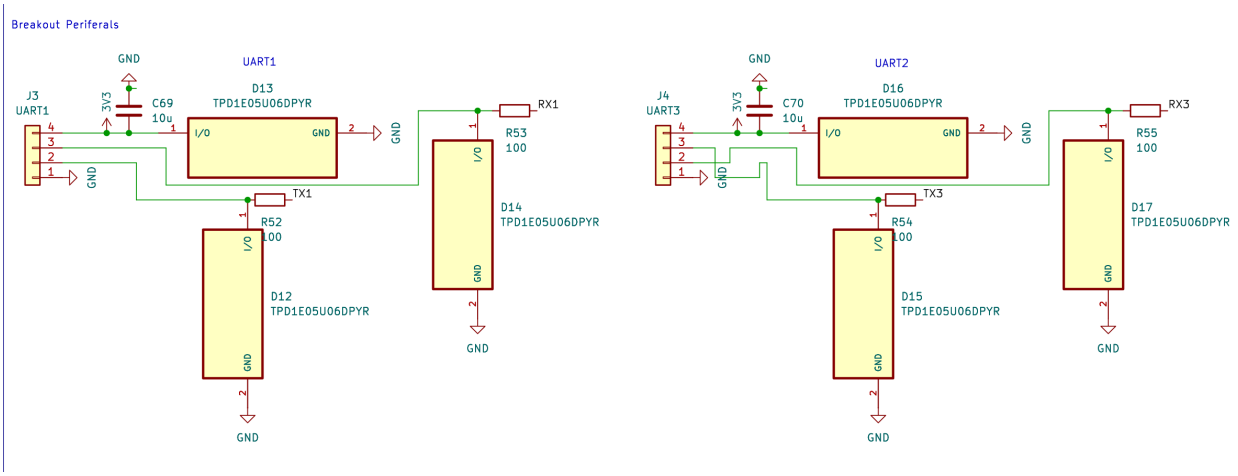
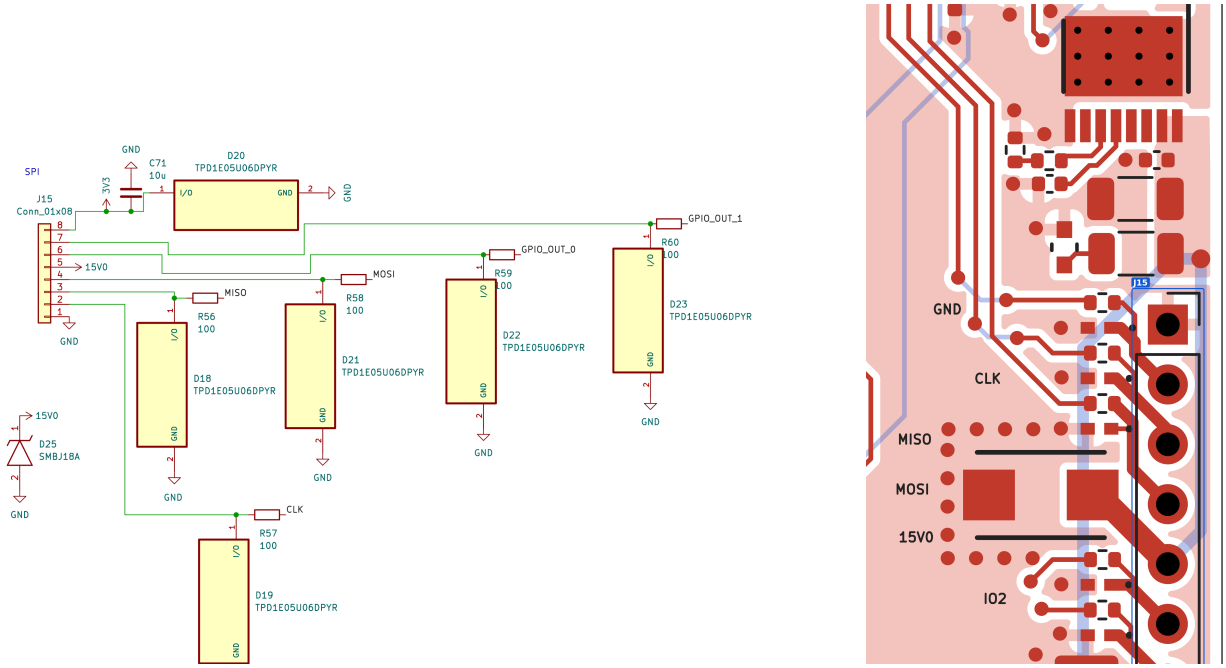


Figure 17: UART breakouts J3 (USART1) and J4 (USART3) with series resistors and ESD diodes.



(a) Schematic **(b)** Layout

Figure 18: 8-pin SPI expansion header J15.

10 PCB layout

Layer	Thickness	Use
F.Cu	35 µm	Components, signal routing, power pours (VBAT, 5 V, 3.3 V, 15 V, switch nodes)
Prepreg	0.1 mm	
In1.Cu	35 µm	Solid GND plane (small VBAT area)
Core	1.24 mm	
In2.Cu	35 µm	3.3 V plane with GND areas, some routing
Prepreg	0.1 mm	
B.Cu	35 µm	Signal routing, GND pour

Table 4: 4-layer stackup (1.6 mm). Board outline 78.7 mm × 71.0 mm, all parts on top, four M3 mounting holes.

- Default signal traces are 0.2 mm; power and off-board signals use 0.5 mm; the battery input uses 1.5 mm. Vias are 0.6 mm/0.3 mm drill (about 450 in total).
- The USB D+/D− pair is 0.27 mm wide, sized with JLCPCB's impedance calculator for this stackup.
- The power stage (XT60, fuse, both switchers and their inductors, the eFuse) is grouped along the top edge, with each converter's input caps placed at its own pins and the switch nodes on small copper pours.
- The TPS55340 and TPS26600 exposed pads have via arrays into the inner ground plane for heat and a low-impedance ground.
- The thermocouple amplifiers and their screw terminals sit along the bottom-left edge and the pressure connectors along the bottom-right edge, at the opposite side of the board from the switchers.

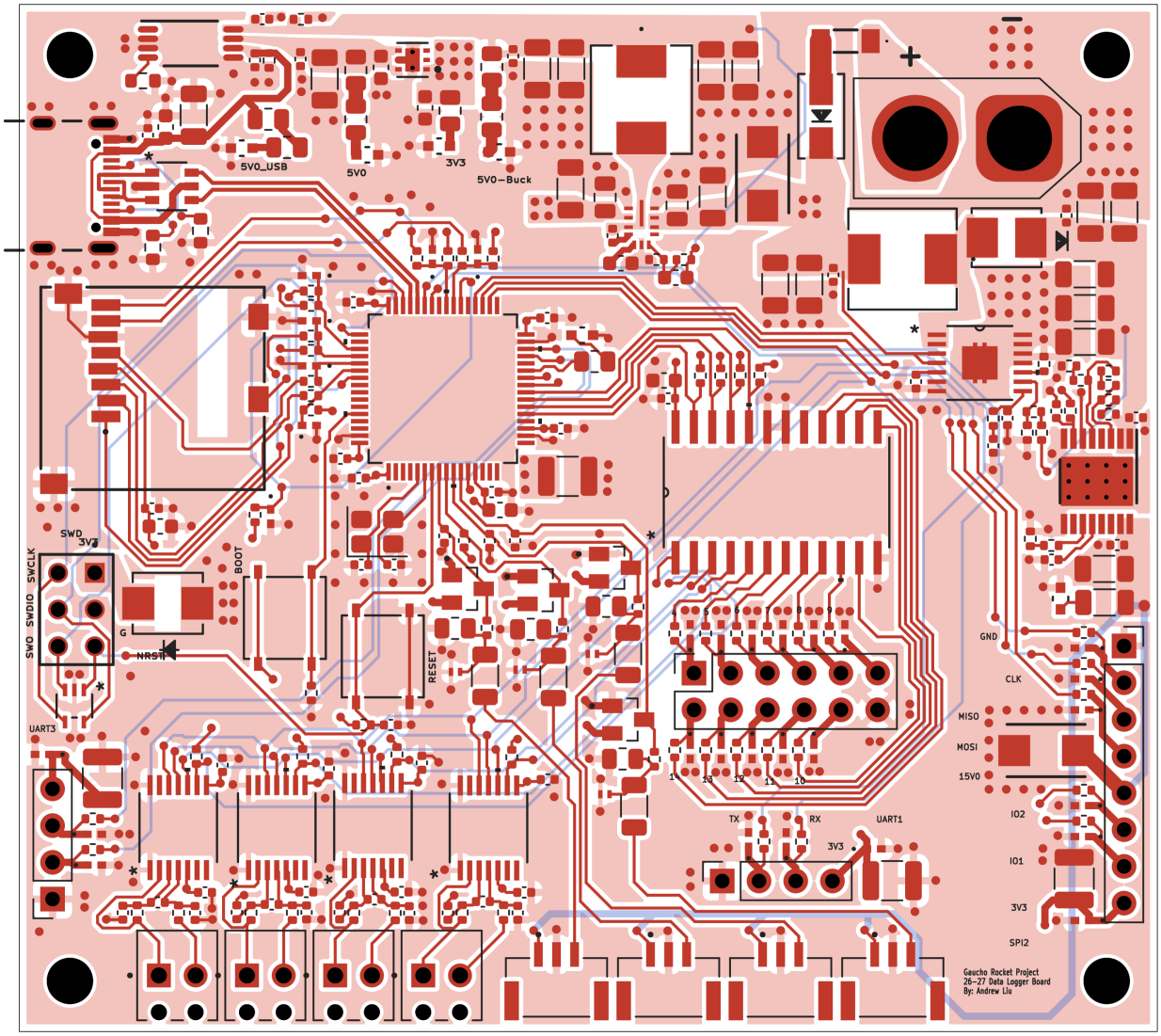
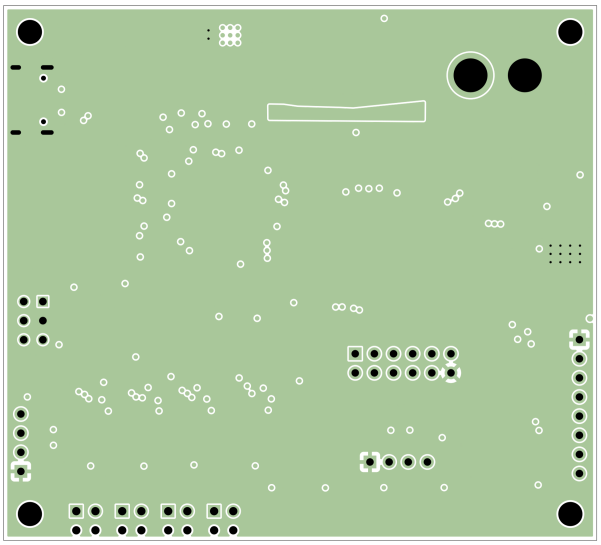
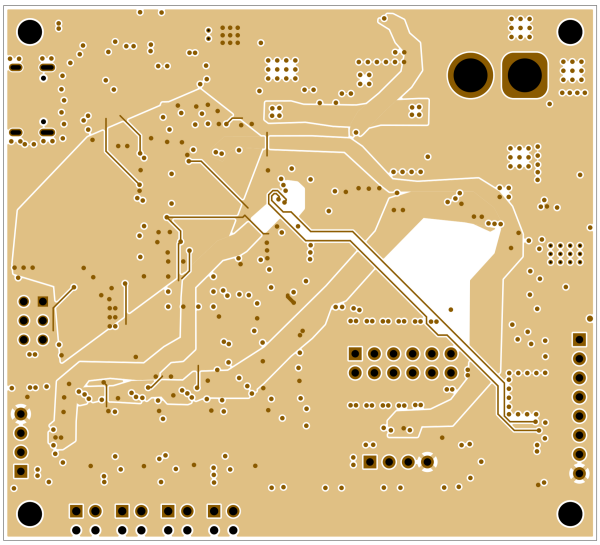


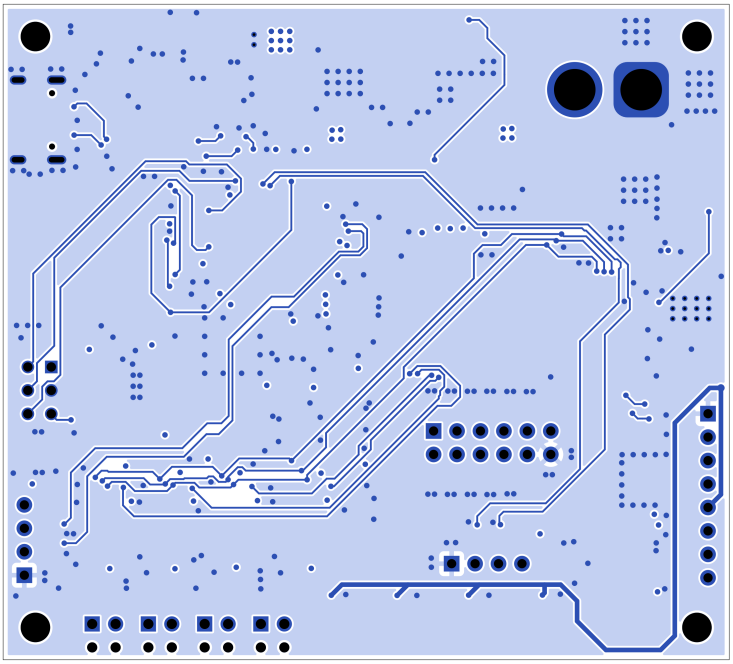
Figure 19: Top copper and silkscreen (bottom copper shown faintly in blue).



(a) In1.Cu (GND)



(b) In2.Cu (3.3 V / GND)



(c) B.Cu (seen from the top)

Figure 20: Inner and bottom copper layers.

11 Differences from the earlier write-up and open items

11.1 Where the schematic differs from the earlier draft or discussion

1. **MCU part number.** The schematic uses the **STM32L496RET6** (512 KB flash); the earlier write-up names the RGT6 (1 MB). The pinout is identical; make sure the BOM orders the one you want.
2. **Decoder enables.** Both $\overline{\text{E1}}$ and $\overline{\text{E2}}$ are grounded and Y15 is used as the idle address (pull-ups on A0–A3). This gives 15 chip selects instead of 16. Driving $\overline{\text{E2}}$ from a GPIO was the other option discussed.
3. **$\overline{\text{SHDN}}$ capacitor.** C78 on the eFuse $\overline{\text{SHDN}}$ pin is 0.1 µF in the schematic (about 10 ms RC); the earlier write-up says 1 µF.

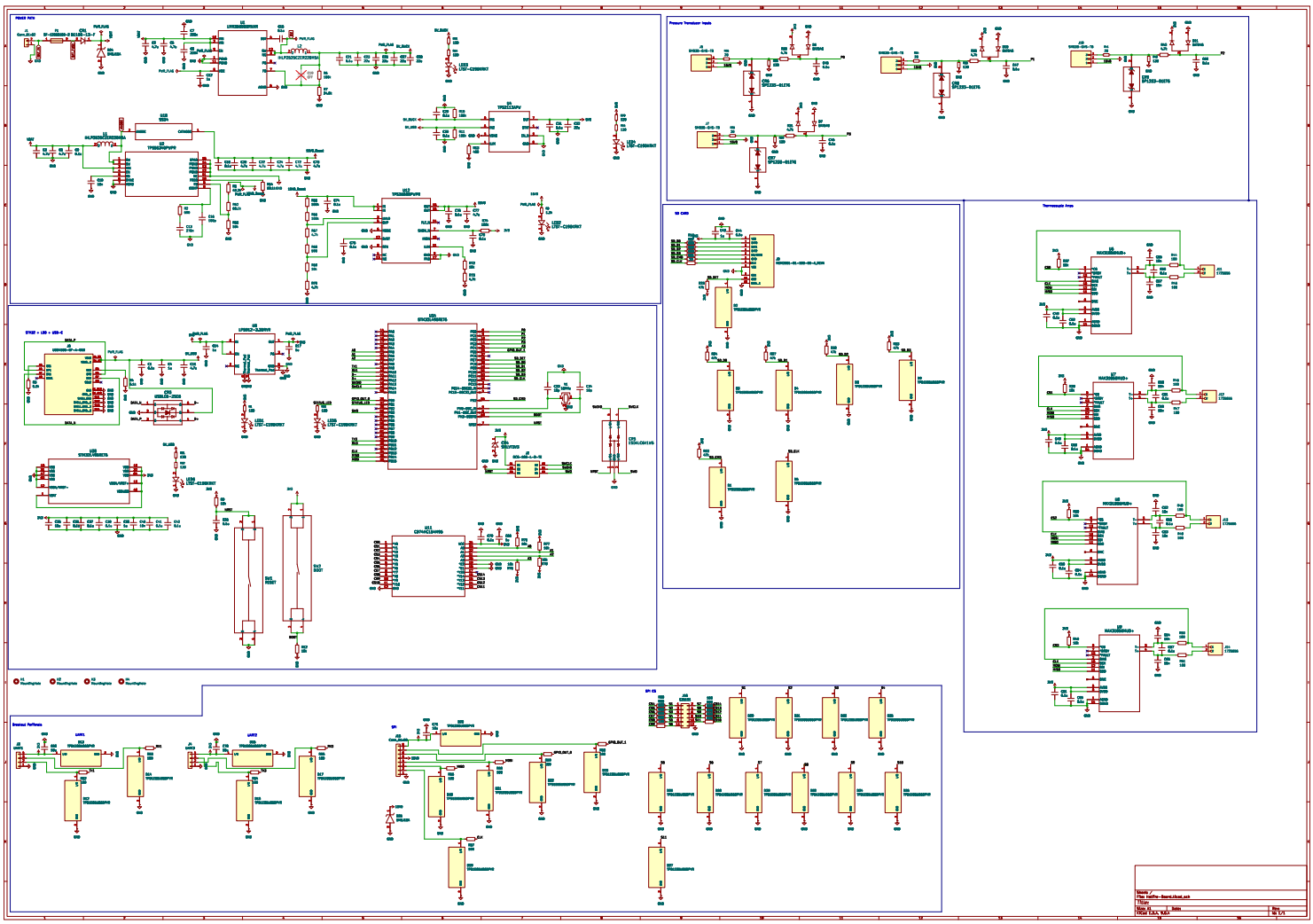
4. **eFuse thresholds.** The earlier write-up gives the UVLO as "12.1–13.09 V" and OVP as "16.49–17.8 V". Those pairs are the falling and rising thresholds (off/on), not min/max.
5. **Superseded sections of the old write-up** (not carried over): the 24 V/flexible battery range, the SMBJ28A TVS, the LMR36006 at 3.3 V (43.2 kΩ), the LMR36006 15 V buck for the transducers, and soldered-wire pressure inputs (now JST GH connectors).
6. **Header label.** J4 is labelled "UART2" in a schematic note but carries USART3 (TX3/RX3). Its TX/RX order is also the reverse of J3 (J3: pin 2 TX; J4: pin 2 RX).
7. **5 V buck inductor.** The schematic uses the Vishay IHLP2525CZER220M5A (22 µH) for both L1 and L2. The Sunlord SWPA6045S220MT was the other part approved; either works.

11.2 Things worth checking before ordering

1. **XT60 footprint.** The schematic still uses KiCad's `AMASS_XT60-F` footprint. Swap it for the Amass XT60PB-M.G.Y footprint (from its LCSC/EasyEDA listing, C19268037, e.g. via easyeda2kicad) and check the pin spacing against the datasheet before placing it.
2. **VDDA filtering.** VDDA (the ADC reference) is fed straight from the 3.3 V rail. A ferrite footprint with 1 µF + 10 nF would allow extra filtering later without a respin.
3. **Transducer cables.** The only way 15 V can reach a pressure signal pin is a wrongly crimped cable, which would overload the 130 Ω 0805 shunt (about 1.3 W). The keyed GH connector and the pin order (15 V, GND, signal) make this unlikely; check each harness before first power-up.

A Full schematic

The complete schematic (A1 sheet, vector) is on the next page; zoom in to read it.


Power Area
Pressure Transducer Inputs
SD Card
Thermocouple Area
STM32 + LSM + ADS-C
Solenoid Platform
SPI
SPI CS
Sheet: /
File: redfin-board.kicad_sch
Title:
Size: A1
Date:
KiCad 9.0.5
Rev:
Id: 1/1